Language Features under Weight Quantization

Activation Agreement and Intervention Effects

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The question

When a model's weights are rounded to fewer bits,

do the internal features that explain it

still do the same job?

27 languages · Gemma-2-2B (main) · Gemma-2-9B confirmation · Llama-3.2-1B cross-check · one fixed SAE per model

Background

- **Quantization** rounds weights to fewer bits (8…4). Reduces cost, but behaviour can change across languages (Marchisio et al., 2024).
- A **sparse autoencoder (SAE)** decomposes a hidden state into a few active **features** from a learned dictionary (Gao et al., 2024).
- Some features fire mostly for **one language**; editing them changes that language's output (Deng et al., 2025).
- Duan (2026) measured whether feature *activations* survive quantization.

We go further: does the feature still do the same thing after quantization?

Related work and our contribution

Work	What they measure	Gap
Marchisio et al. (2024)	Task outputs across languages after quantization	No internal mechanism
Deng et al. (2025)	Language-selective SAE features; causal edits	No quantization
Duan (2026)	Activation survival (R_1) under quantization	No functional test
Makelov et al. (2025)	Judge features by edits, not proxy metrics	Not applied to quantization

We connect these: measure both activation survival and functional preservation, and show they come apart.

Research questions

RQ1 Which language features retain their **activation patterns** after quantization?

RQ2 How does **editing** language features affect prediction and comprehension before and after quantization?

RQ3 How well do **activation measures** predict whether a feature's **output effect** is preserved?

RQ4 How much does the **same feature edit** change when the model is quantized, and what explains that change?

Experimental setup

Models & SAEs

- Gemma-2-2B, layer-12, 16k JumpReLU SAE
- Gemma-2-9B, layer-20, 16k SAE
- Llama-3.2-1B, layer-12, TopK SAE (k=48)

Data

- 27 FLORES-200 languages (9 per resource tier)
- 300 aligned devtest sentences per language
- Belebele: 385 questions $\times$ 18 languages

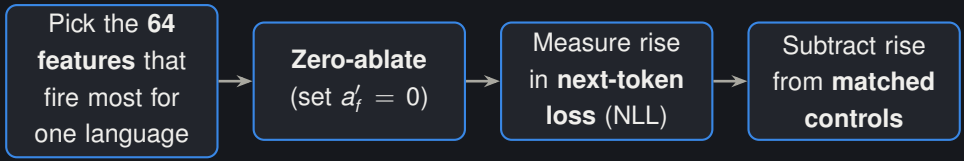
Quantizers

- Legacy RTN (reproducing Duan)
- Matched RTN (one scale per channel)
- GPTQ (calibrated, 221k tokens)

Feature selection

- 8,086 language-selective features (2B)
- Top K=64 per language (“K64 group”)
- Activity-matched control groups

Methodology: the core test



Excess harm: $E^b_\ell = \Delta\text{NLL}(\text{group}_\ell) - \overline{\Delta\text{NLL}(\text{controls}_\ell)}$, Effect ratio: $\rho = E^{(b)}/E^{(\text{FP})}$

Repeat at FP and with weights rounded to **8, 7, 6, 5, 4 bits**

A correction to earlier work (Duan, 2026)

	No BOS	With BOS
FP perplexity	458.4	16.1
INT6 survival	51%	97.5%
INT7 survival	81%	$>$99.9%

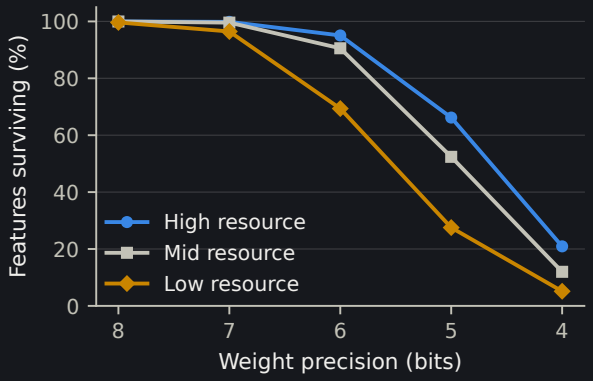
We reproduced Duan's numbers with their code (INT6: 51.15% vs. 51.30%).

The gap: their Gemma inputs omitted the **beginning-of-sequence (BOS) token**.

One token shifted perplexity 28$\times$ and halved apparent feature survival.

All subsequent results use corrected inputs. We also retired our own first pipeline after finding separate flaws.

RQ1: Which activations survive?



Activation survival:

$$R_1 = \mathrm{corr}_t\left(a_f^{\mathrm{FP}}(t),\, a_f^{b}(t)\right)$$

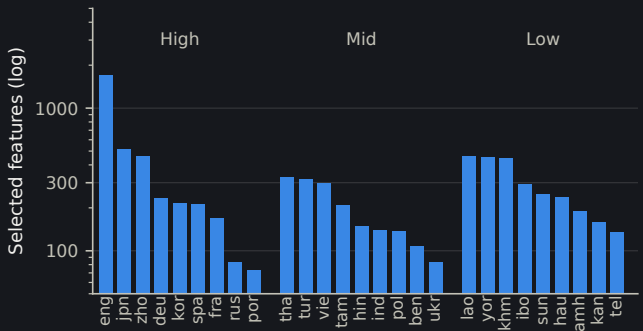
Rule: $R_1 > 0.9$.

Low-resource languages lose more features; selective features survive slightly less often.

Both gaps trace to weaker activations ($\rho_s = 0.35$ at INT6). Matching on strength removes the gap.

Selective features: median peak activation 9.8 vs. controls 14.2 at $S > 5$.

RQ1: Feature counts do not follow resource tier



Range: 73 (Portuguese) to 1,719 (English).

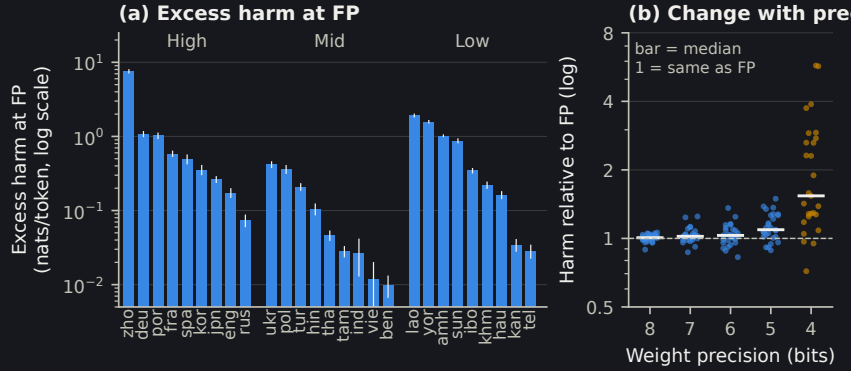
Lao (464), Yoruba (453), Khmer (444) exceed most high-resource languages.

Close relatives share: Portuguese–Spanish, Russian (83)–Ukrainian (84).

$\rho_s = -0.07$ (count vs. tier).

Count reflects how distinct a language looks to the model, not its resources (descriptive, post hoc).

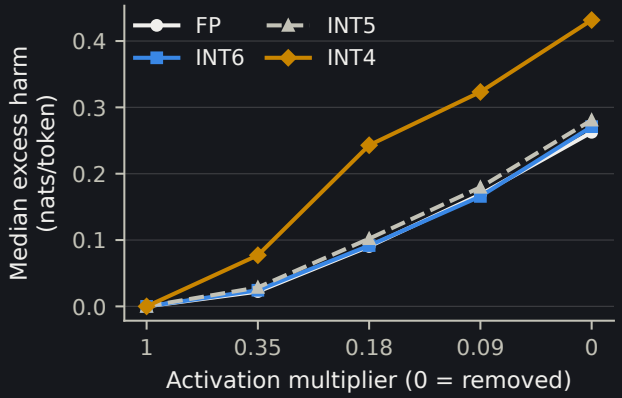
RQ2: Language features matter in every language



Left: FP excess harm positive in all 27 languages. Mean 0.704 [0.684, 0.723], median 0.262 nats/token.

Right: INT4 median ratio $\rho = 1.54$ [1.43, 1.62]; mean ratio 2.14 [2.04, 2.50].

RQ2: Dose–response confirms causal role



Gradual suppression: $a'_f = \alpha\, a_f$.

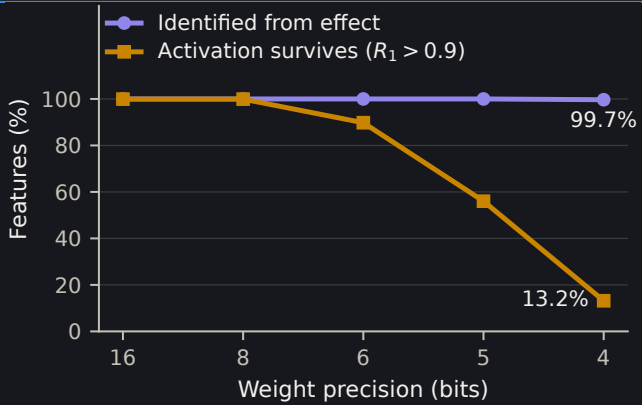
FP harm increases monotonically in all 27 languages.

Median harm:
$0.022 \rightarrow 0.090 \rightarrow 0.169 \rightarrow 0.262$ nats/token.

INT4 curve is higher at every step.

Dose and consistency rules hold; strict precision-wise monotonicity fails due to a small intermediate-width dip.

RQ3: Activation survival $\neq$ functional preservation



Effect fingerprint cosine:

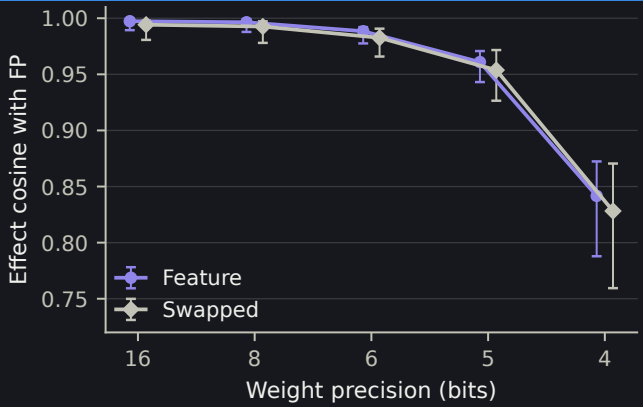
$$W_1 = \frac{\mathbf{d}_f^{\mathrm{FP}} \cdot \mathbf{d}_f^{b}}{\|\mathbf{d}_f^{\mathrm{FP}}\| \, \|\mathbf{d}_f^{b}\|}$$

At legacy INT4 ($n = 1{,}487$):

$R_1 > 0.9$ survival	13.2%
Identified by W_1	99.7%
Median W_1	0.839

Activation cutoff declares 87% “dead”; almost all still produce recognizable output effects.

RQ3: The 4-bit drift is generic (direction-swap control)



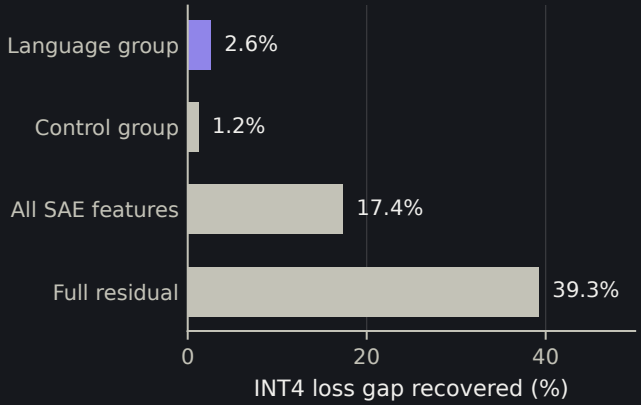
Ablate a **different feature's direction** at the same positions and strengths.

Feature direction W_1	0.842
Swapped direction W_1	0.828
Gap	0.014

Gap is small and consistent at every precision.

INT4 similarity loss is generic to downstream computation, not specific to the features.

RQ3: Restoring features recovers almost nothing



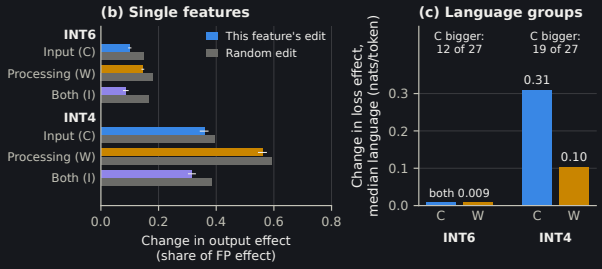
Patch FP activations into INT4:

Language group K64	2.6% [2.4, 2
Matched control	1.2
Full residual	39.3% [38.5, 40

GPTQ: 2.0%.

Language-group features carry $<$3% of the INT4 compression damage.

RQ4: Decomposing the change in a fixed edit



Fix edit at FP direction and strength.

Cross FP/quantized layers:

$$C = D[\mathrm{Q}, \mathrm{FP}] - D[\mathrm{FP}, \mathrm{FP}]$$

$$W = D[\mathrm{FP}, \mathrm{Q}] - D[\mathrm{FP}, \mathrm{FP}]$$

$$I = D[\mathrm{Q}, \mathrm{Q}] - C - W - D[\mathrm{FP}, \mathrm{FP}]$$

At INT4: median $|C| = 0.310$, $|W| = 0.104$ nats/token.

For single-feature output vectors, W is larger. Attribution depends on the edit and the outcome.

The Belebele null (our only confirmatory test)

Condition	Accuracy
Unedited	42.6%
K64 removed	42.6%
Control removed	42.5%
FP excess loss	-0.0001 nats
95% CI	$[-0.0035, +0.0031]$

This was our held-out, pre-registered test.

Zeroing K64 does not change Belebele reading-comprehension accuracy.

Larger edits (1,024–2,860 features) *do* lower accuracy by 1.2–6.0 pts. Sham edits change nothing.

The groups help next-token prediction but have no detectable effect on this comprehension task.

Cross-model and cross-quantizer checks

Gemma-2-9B

FP excess harm positive in 27/27 (median 0.270 $[0.197, 0.608]$).

INT4 ratio 1.586 $[1.238, 2.264]$.

All frozen predictions met. Two defective criteria documented.

Three quantizers

Legacy, matched RTN and GPTQ give the same qualitative pattern.

GPTQ halves INT4 amplification (ratio 1.22 vs. 1.54 on 2B).

Llama-3.2-1B

FP effect in 21/27 (rule: 24). Verdict: not consistent.

Median $E = 0.048$; INT4 ratio 0.86.

Model, SAE, tokenizer all differ.

Summary: INT4 measurements across quantizers (Gemma-2-2B)

INT4 measurement	Legacy	Matched	GPTQ
Active-feature survival (%)	12.7	18.6	30.5
Mean loss increase (nats/token)	0.590	0.385	0.130
Median group-effect ratio (ρ)	1.54	1.36	1.22
Median effect cosine (W_1)	0.839	0.875	0.923
Transfer cosine	0.768	0.822	0.877
Belebele accuracy (%)	23.6	29.1	31.8

Rows answer different

questions. Survival: active-feature shares per language; W_1: F5 sample (1,487 features); transfer: eligible feature–context pairs; Belebele: 18 languages.

Limitations

- **One layer, one SAE per model:** language-specific neurons concentrate in early/late layers (Tang et al., 2024; Kojima et al., 2024); middle-layer test is conservative.
- **Fixed-length inputs:** truncation removes a length confound but may change internal representations. Paired comparisons keep both sides equal.
- **Mostly exploratory:** most FLORES analyses reuse examined sentences. Only Belebele was confirmatory (null result).
- **Llama inconsistency:** cross-family check failed (21/27), but model, SAE, tokenizer all changed.

Claims

1. Language features contribute to next-token prediction in all 27 languages on both Gemma sizes, with dose–response in every language.
2. The activation-survival cutoff understates functional preservation: 13.2% pass $R_1 > 0.9$, yet 99.7% are identifiable by output effect W_1.
3. The INT4 effect-similarity loss is generic (direction-swap gap 0.014), and restoring language-group activations recovers only 2.6% $[2.4, 2.8]$ of the loss gap.
4. Activation agreement, task performance and an edit's output effect require separate measurements: they answer different questions and give different answers.

Reproducibility

- Reproduced Duan's numbers with their released code before reporting the BOS issue
- 9B predictions and code hashes frozen before any 9B data existed
- Two defective 9B prediction criteria documented, not counted
- Null results (Belebele, Llama, failed repair) reported with same detail as positives

`github.com/TH4KUR/always-late`
`huggingface.co/datasets/karma-skz/always-late-artifacts`
`wandb.ai/karma-skz-/anlp-project`